Data Technician Week 5 Workbook Cloud Computing and Azure Data Fundamentals

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Day 1: Task 1

Please research and complete the below questions relating to key concepts of cloud.

Be prepared to discuss the below in the group following this task.

What can cloud computing do for us in the real-world?	Cloud computing allows us to rent computing power, storage, and software over the internet from providers such as Microsoft Azure, AWS, or google cloud, rather than buying and maintaining physical servers on-site. It can be accessed anywhere at any time, just needing internet connection. Data protection is improved as cloud services and vendors can offer different security features for recovery and backup, meaning personal information could be safer than if you lose a physical device with data such as USB.
How can it benefit a business?	• **Cost Savings:** Companies shift from massive upfront capital expenses (buying hardware) to predictable operating expenses (paying only for what they use). • **Scalability:** Businesses can instantly add more server capacity during high-traffic events—like Black Friday sales—and scale back down afterward to avoid paying for idle resources. • **Global Access:** Employees can securely log in to company software, databases, and collaboration tools from any device or location, which directly supports remote and hybrid work. • **Business Continuity:** Data is automatically backed up across multiple secure data centers, protecting the company from losing vital information during power outages, cyberattacks, or natural disasters. • **Automatic Updates:** Cloud providers handle all software patches, security upgrades, and system maintenance, freeing up internal IT teams to focus on core projects. • **Faster Innovation:** Developers can deploy new applications and test digital products in hours rather than the weeks it would take to order and configure physical hardware.

What's the alternative to cloud computing?	**The 3 Main Alternatives to Public Cloud** • **On-Premises Infrastructure:** You buy physical servers, networking switches, and storage arrays, putting them in a dedicated server room or closet on-site. Your internal IT staff handles all security, cooling, electricity, and maintenance. • **Colocation (Data Center Leasing):** Instead of keeping servers in your office, you rent physical space, power, and cooling in a third-party commercial data center. You still own and maintain the physical servers, but the facility provider keeps the power running and the building secure. • **Private Cloud:** This is a dedicated cloud environment built exclusively for one business. It can be hosted on your own premises or at a colocation facility. It gives you cloud-like flexibility and automation, but the resources are not shared with any other company.
What cloud providers can we use, what are their features and functions?	AWS offers the widest range of cloud services, Azure integrates well with Microsoft's ecosystem, such as Microsoft 365, and google cloud is known for its strengths in data analytics, AI, and machine learning. These platforms help businesses reduce IT costs and quickly scale resources.

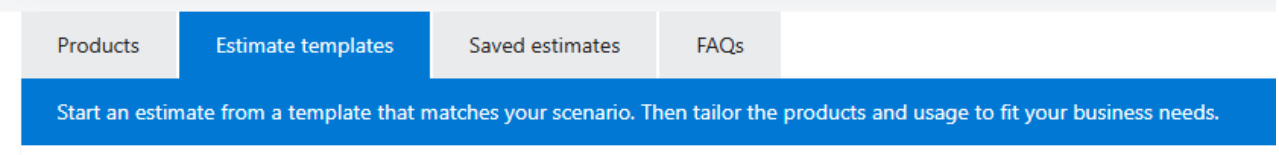
Day 1: Task 2

Read the information on this page: [Explore the pricing calculator - Training | Microsoft Learn]

Activity: Explore different scenarios

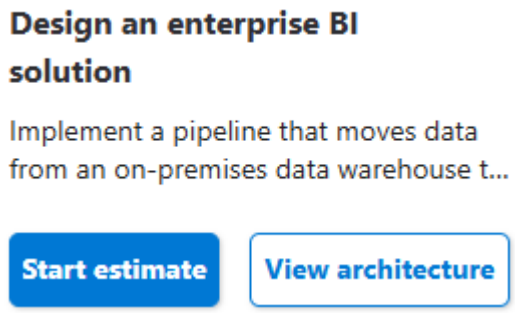
1) Visit this page [Pricing Calculator | Microsoft Azure]

2) Choose **Estimate templates**



3) Start estimate for **Design an enterprise BI solution**

Implement a pipeline that moves data from an on-premises data warehouse to a cloud and then use a business intelligence (BI) model to serve the data.



4) Explore the services and the cost.

Cost of a Design an enterprise BI solution	$7799

Day 1: Task 3

Please research the below cloud offerings, explain what they are and examples of use cases.

Cloud Offerings	Explain what it is	When / how might you use this service in the real-world?
IaaS (Infrastructure as a service)	It provides the fundamental building blocks of computing over the internet. Instead of buying physical hardware, you rent raw virtual servers, storage space, and networking infrastructure from a cloud provider. You are responsible for managing the operating system, apps, and data.	• Hosting a massive, high-traffic website or e-commerce platform that needs to scale its server capacity up and down instantly. • Backing up enterprise company data or running complex simulations that require massive raw processing power.

PaaS (Platform as a service)	It provides a pre-configured hardware and software environment specifically designed for building and deploying applications. The cloud provider handles the underlying servers, operating systems, and storage, leaving developers free to focus entirely on writing code.	○ A software development team building a new mobile app or web service from scratch who wants to write code without spending weeks setting up databases or server environments. ○ Analyzing big data sets using built-in, ready-to-use business intelligence tools.
SaaS (Software as a service)	It delivers a complete, fully functional software application directly over the internet, typically accessed through a web browser or mobile app. The provider manages absolutely everything—from the code to the physical hardware.	○ Managing day-to-day business operations using web-based tools like Google Workspace, Microsoft 365, or Salesforce. ○ Everyday consumer activities like streaming movies on Netflix, listening to music on Spotify, or sending team messages on Slack.

Day 1: Task 4

Please research the below terms and explain what they are, when they would be appropriate and a real-world example of where it could be implemented (i.e. what type of organisation).

Public Cloud	• **What it is:** A computing model where infrastructure, storage, and applications are owned and operated by a third-party provider (like AWS or Microsoft Azure) and shared across multiple organizations over the public internet. • **When it is appropriate:** When an organization needs to deploy applications rapidly, wants to avoid upfront hardware costs, or requires the ability to scale resources up and down instantly based on unpredictable traffic. • **Real-world organizational example:** A fast-growing tech startup or an e-commerce retailer. For example, an online clothing brand uses the public cloud to run its website so it can seamlessly handle a massive spike in users during Holiday sales without buying extra physical servers that would sit idle the rest of the year.
Private Cloud	• **What it is:** A cloud environment dedicated exclusively to a single organization. It can be physically located at the company's on-site data center or hosted by a third party, but the hardware and network are completely isolated from other users. • **When it is appropriate:** When an organization handles extremely sensitive data, must comply with strict data privacy regulations, and requires total control over its security configurations and hardware performance. • **Real-world organizational example:** A commercial bank or a large healthcare provider. A national bank utilizes a private cloud to store customer financial records and process transactions, ensuring absolute data isolation to prevent data breaches and meet strict government banking regulations.
Hybrid Cloud	• **What it is:** A blended environment that combines public and private clouds, allowing data and applications to be shared and moved smoothly between them. • **When it is appropriate:** When a business wants to keep its core, sensitive data locked down in a highly secure environment but still wants to leverage the massive computing power and cost-efficiency of the public cloud for less critical workloads. • **Real-world organizational example:** A large university or a legacy manufacturing corporation. A

	university might keep its sensitive student records and financial payroll safely locked in an on-premise private cloud, while using a public cloud to host its public-facing website and student learning management platforms (like Canvas or Blackboard).
Community Clou	• **What it is:** A collaborative cloud infrastructure shared by several organizations that have similar compliance, security, or project goals. It is managed either by the organizations themselves or a third party. • **When it is appropriate:** When multiple independent entities need to work together on shared projects, share specialized tools, or meet identical regulatory standards, allowing them to split the infrastructure costs. • **Real-world organizational example:** A regional police department coalition or a group of government agencies. For instance, several local police forces and emergency services might share a community cloud to securely access a joint criminal database, ensuring they all meet the same strict law-enforcement security standards while splitting the operational costs.

Day 2: Task 1

You are a Cloud Consultant hired by a client. You will identify the most suitable cloud service model for a client requiring control over infrastructure, estimate potential cloud costs, and explain the benefits of cloud adoption.

🏭 Client – "DataCore Enterprises"

- A large enterprise wants to move its internal database servers to the cloud.
- They already have IT staff.
- They want full control over operating systems and storage configuration.

Questions for you:

- Which service model suits them? (IaaS, PaaS, Saas)
- How can they estimate the cost of using cloud services?
- What are the possible benefits of using cloud services?
- Bonus: What might be potential risks?

Day 2: Task 2

Describe, with examples, the **three** major areas that the Computer Misuse Act deals with.

Area	Description	Example
Unauthorised access to computer material	Gaining access to a computer system, its programs, or its data without having the owner's explicit permission. This covers any form of basic intentional breaking-in or "snooping", even if no harm is done or nothing is stolen or changed.	Logged into a classmate's or colleague's online account using a password you guessed or secretly copied without their permission just to read their private messages.
Unauthorised access with intent to commit further offences	Gaining unauthorised access to a computer system or network specifically to prepare for or carry out a secondary, more serious criminal act (such as theft, fraud or blackmail).	Hacking into a retail company's secure backend database with the clear, direct intention of stealing customer credit card details to buy goods illegally.
Unauthorised modification of computer material	Unlawfully altering, destroying, erasing, or imparing any data or software stored on a computer system without permission. This also includes creating, obtaining, or intentionally distributing malware to distrupt system operations.	Writing a piece of ransomware, or spreading a virus across a school network to corrupt files, change grades or deliberately crash the servers.

The computer misuse act 1990 is an act where an individual can be criminalised because of computer related offense. Describe three extra powers that the Police and Justice Act 2006 (Computer Misuse) has added.

Description
1. Criminalising "Hacking Tools" and Malware Distribution (Section 3A): The amendment made it a specific crime to make, adapt, supply, or obtain "articles" (such as specialized hacking software, malware, or lists of stolen passwords) knowing or believing they are likely to be used to commit a cyber offence.
2. Explicitly Outlawing Denial-of-Service (DoS) Attacks: The original Section 3 ("unauthorised modification of computer material") was completely reworded to focus on "unauthorised acts with intent to impair, or with recklessness as to impairing, the

operation of a computer." This modification closed a legal loophole, making Distributed Denial of Service (DDoS) attacks that flood and crash network servers explicitly illegal.
3. Substantially Increasing Maximum Prison Sentences: The act significantly increased the tariffs and penalties for offences. The maximum sentence for basic unauthorised access (hacking) rose from 6 months to 2 years imprisonment, while the maximum sentence for intentionally impairing or damaging a system rose to 10 years.

Look at the below website to answer the questions:
https://www.gov.uk/personal-data-my-employer-can-keep-about-me

Write down three items of data which a company can store about an employee.
Personal identification: full legal name and home address.
Employment records: National insurance (NI) number (required for UK tax and employment verification.
Financial details: Bank account number and sort code (used for payroll processing).

Give three more examples of data that an employer can only store if they first get the employee's permission.
1. Public Marketing Media (e.g., Photographs and Videos): Storing an employee's headshot, promotional photos, or video interviews to use on the public company website, corporate social media channels, or external recruitment brochures.
2. Biometric Identification Data: Storing an employee's fingerprints or facial recognition geometry to manage building access or digital clock-in systems, provided a non-biometric alternative (like a standard swipe card) is available.
3. Voluntary Corporate Benefits & Wellness Data: Storing and sharing information needed to register an employee for non-mandatory third-party perks, such as voluntary health-tracking apps, corporate gym memberships, or private retail discount schemes.

Conduct further research to answer the below questions.

Question	Answer
Provide one example of: Copyright infringement	Downloading a commercial movie or music album from an unauthorised torent website and sharing it with others without purchasing a license.
Provide one example of: Plagiarism	Copying and pasting text directs from an online article into a school essay and submitting it as your own work without citing the author.

What are two consequences of copyright infringement and software piracy?	Financial loss for creators: software developers, musicians, and companies lose revenue, which harms the industry and reduces future innovation. Legal action and heavy fines: copyright holders can sue individuals or businesses for damage, which can lead to court-ordered financial penalties.
Give three possible consequences for individuals when using pirated software	Malware and security infections: Cracks and Keygens frequently bundle hidden spyware, ransomware, or trojans that compromise personal data. Lack of software updates: The program cannot access official security patches or bug fixes, leaving the system highly vulnerable to exploits. Loss of technical support: software vendors will refuse to assisst you if the software breaks, malfunctions, or corrupts your existing files.

Listed below are some laws which we have covered today:

1. Computer Misuse Act 1990
2. Police and Justice Act 2006 (Computer Misuse)
3. Copyright, Designs and Patents Act 1988
4. Copyright (Computer Programs) Regulations 1992
5. The Health and Safety (Display Screen Equipment) Regulations 1992
6. Data Protection Act 2018
7. Consumer Rights Act 2015

- Insert a number in the first column of each row to match each of the statements with one of the above Acts.

- One of statements is incorrect and not illegal. For this statement, write 'Not illegal'.

Act number	Clause
4	With some exceptions, it is illegal to use unlicensed software
7	Any product, digital or otherwise, must be fit for the purpose it is supplied for
1	Unauthorised modification of computer material is illegal
Not illegal	It is illegal to create or use a hacking tool for penetration testing

6	Personal data may only be used for specified, explicit purposes
5	Employers must provide their computer users with adequate health and safety training for any workstation they work at
2	It is illegal to distribute hacking tools for criminal purposes
3	It is illegal to distribute an illicit recording
6	Personal data may not be kept longer than necessary
1	Gaining unauthorised access to a computer system is illegal
5	Employers must ensure that employees take regular and adequate breaks from looking at their screens
2	It is illegal to prevent or hinder access (e.g. by a denial-of-service attack) to any program or data held in any computer
6	Personal data must be accurate and where necessary kept up to date

Day 2: Task 3 (if finished Day 2 Task 2 early)

Please complete this activity to the best of your ability. This is an introduction to Azure Cloud Computing. You can skip Exercises as we will complete day later. Sing in using bootcamp email to save your progress.
Introduction to Cloud Infrastructure: Describe Azure Architecture and Services - Training | Microsoft Learn

Screenshot of XP

Day 3: Task 1: Activity: Azure Services

Research **3** of the following Azure Services and write a brief description of their main points.

Use Azure products website or search online: [Directory of Azure Cloud Services | Microsoft Azure]

- API Management
- Application Insights
- Azure Active Directory
- Azure Rights Management
- Batch
- BizTalk Services
- Data Factory
- HDInsight
- StorSimple
- Operational Insights

Service 1 -

API Management

- **Purpose:** A platform for publishing, managing, securing, and analyzing application programming interfaces (**APIs**).
- **Key Functions:** Acts as an API gateway that controls traffic, handles **authentication/security tokens**, enforces rate limits, and provides a developer portal for API documentation.

Service 2 -

Application Insights

- **Purpose:** An extension of Azure Monitor used for managing and monitoring **application performance** (APM).
- **Key Functions:** Monitors live web applications to automatically detect performance anomalies, log application errors, track user behaviour metrics, and help developers diagnose bugs.

Service 3 -

Data Factory

- **Purpose:** A cloud-based **data integration service** used to automate data workflows.
- **Key Functions:** Allows developers to create data-driven workflows (pipelines) to orchestrate and automate data movement and data transformation (**ETL/ELT**) between different cloud or

on-premises storage systems.

Challenge: Explain if any other Azure Cloud service might benefit you or your business (optional).

Service -

Azure Simulators

Learn by doing it! Each simulation below links out to an interactive lab that you can complete, just as if you were in a real Azure environment! Access any simulation you wish, as often as you wish.

- **CREATE A RESOURCE GROUP**
  In this simulation, you will be guided, step-by-step, through the process of creating a new resource group in Microsoft Azure, using the Azure Portal.

- **DEPLOY A WINDOWS VM**
  In this simulation, you will be guided, step-by-step, through the process of provisioning a new Windows virtual machine in Microsoft Azure, via the Azure Portal.

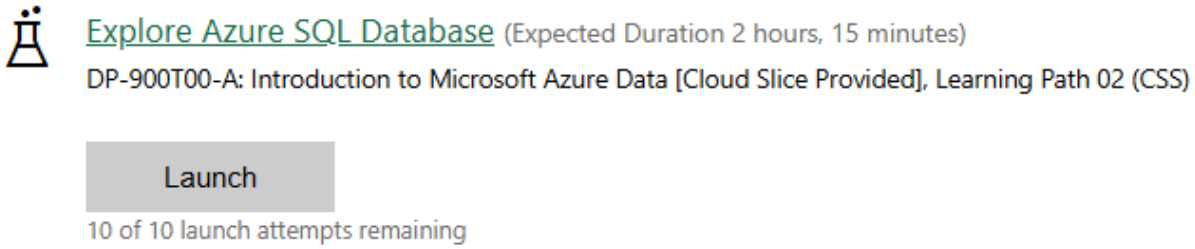
- **PROVISION A STORAGE ACCOUNT**
  In this simulation, you will be guided, step-by-step, through the process of provisioning a new General Purpose V2 storage account in Azure, using the Azure Portal.

- **DEPLOY AN AZURE FILE SHARE**
  In this simulation, you will be guided through the process of creating an Azure file share, called **myfileshare** in an existing storage account.

Day 3: Task 2: Skillable

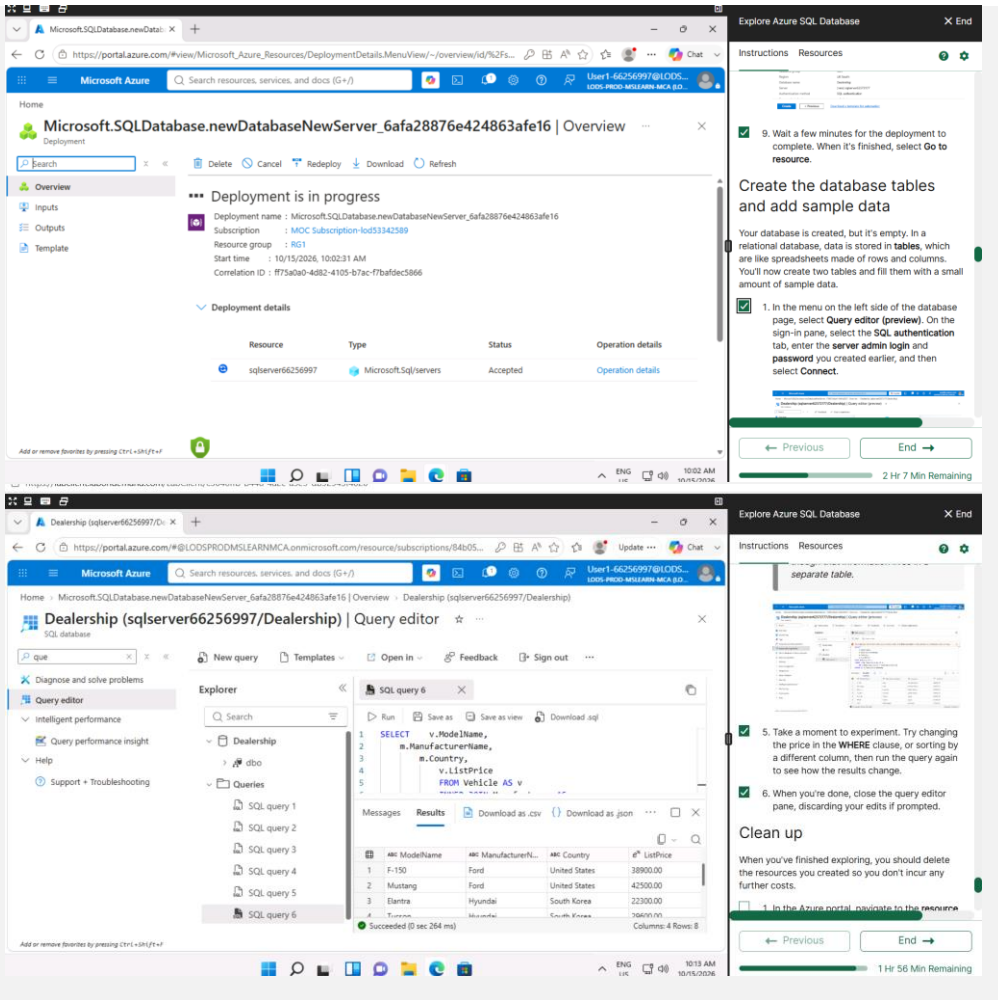
Please complete the below lab (1***) Explore Azure SQL Database*** and paste evidence of the completed lab in the box provided.

1) Visit https://leeptalent.learnondemand.net/
2) Sign into your account
3) Select activity **DP-900T00-A Introduction to Microsoft Azure Data**
4) Find and Launch **Explore Azure SQL Database**



You can watch this video on how to complete the lab.
Lab 1 Explore Azure SQL Database.webm

Completed Lab



Day 3: Task 3: Skillable

Please complete the below lab (2) ***Explore Azure Storage for non-relational data*** and paste evidence of the completed lab in the box provided.

1) Visit https://leeptalent.learnondemand.net/
2) Sign into your account
5) Select activity **DP-900T00-A Introduction to Microsoft Azure Data**

3) Find and Launch **Explore Azure Storage for non-relational data.**



Explore Azure Storage for non-relational data (Expected Duration 2 hours, 15 minutes)

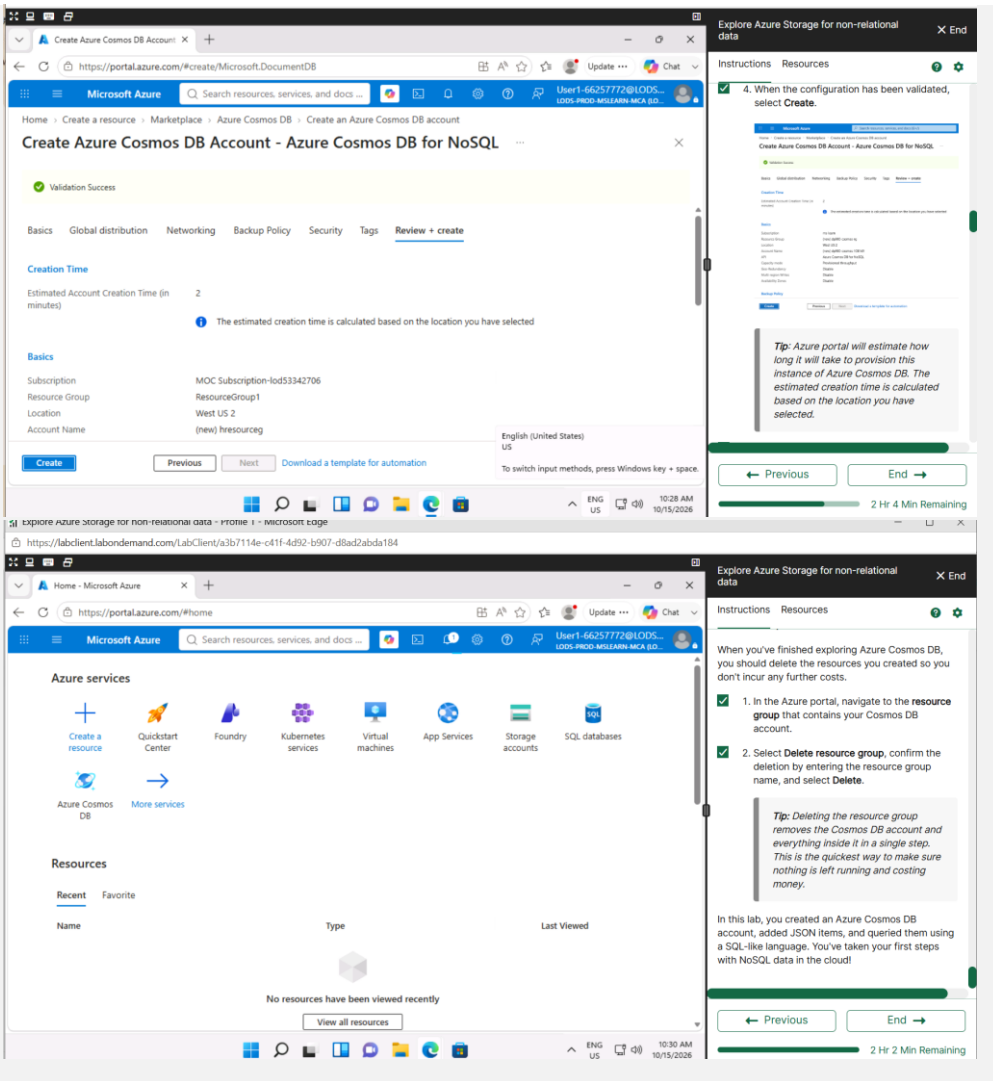
DP-900T00-A: Introduction to Microsoft Azure Data [Cloud Slice Provided], Learning Path 03 (CSS)

Launch

10 of 10 launch attempts remaining

You can watch this video tutorials on how to complete the labs.
Part 1: Explore Storage Account: Azure Lab 2 Storage Account.mp4
Part 2: Explore Azure Cosmos DB Azure Lab 2 Azure Cosmos DB.webm

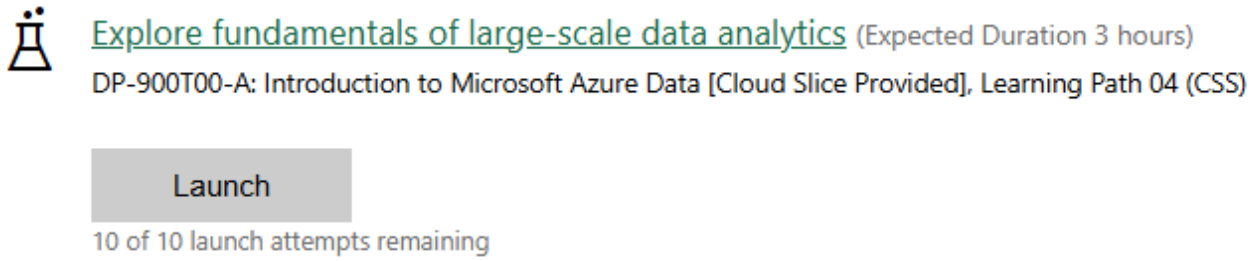
Completed Lab



Day 4: Task 1: Skillable

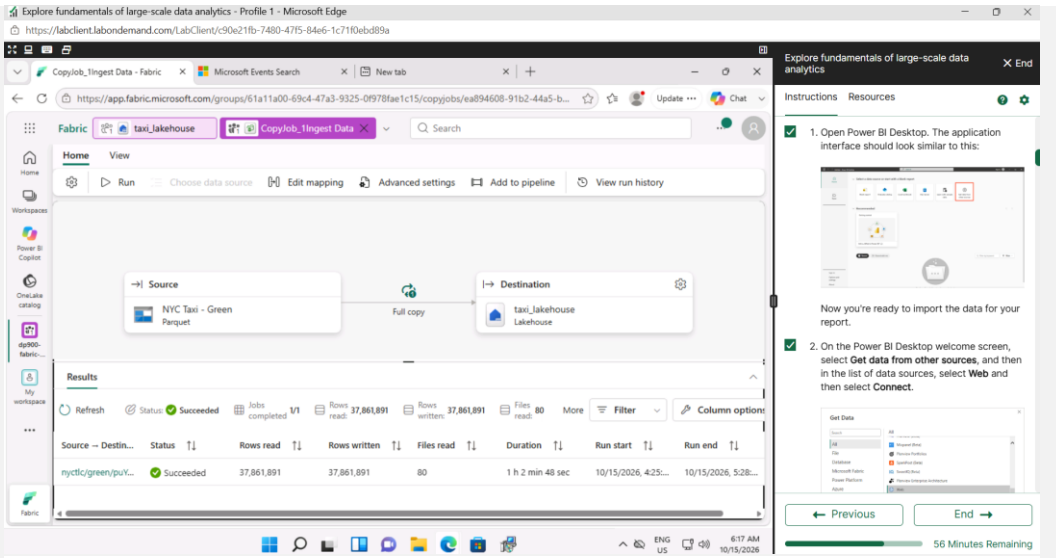
Please complete the below lab (3) **Explore fundamentals of large-scale data analytics** and paste evidence of the completed lab in the box provided.

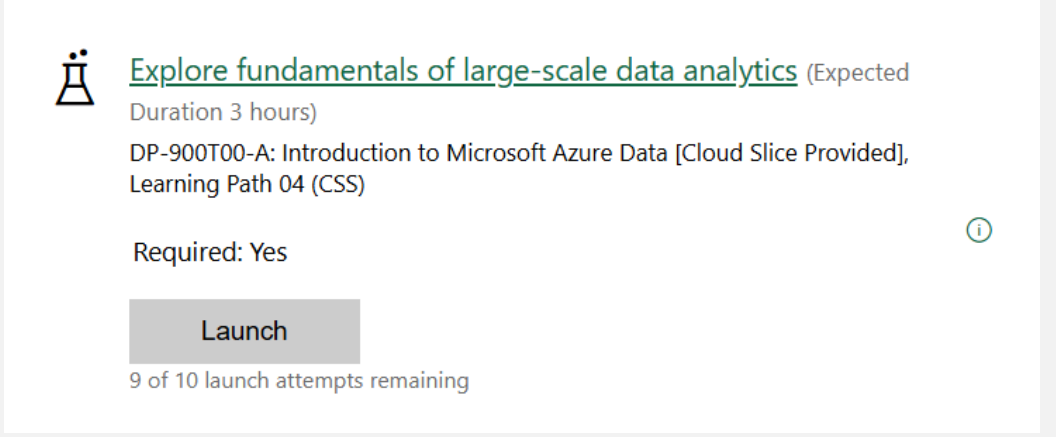
1) Visit https://leeptalent.learnondemand.net/
2) Sign into your account
3) Select activity **DP-900T00-A Introduction to Microsoft Azure Data**
4) Find and Launch **Explore Azure Storage for non-relational data Explore fundamentals of large-scale data analytics**



You can watch this video on how to complete the lab.
Part 1: Azure Lab 3 Part 1 Explore data analytics in Microsoft Fabric.webm
Part 2: Azure Lab 3 Part 2 Explore real-time analytics in Microsoft Fabric.webm
Part 3: Azure Lab 3 Part 3 Explore fundamentals of data visualization Power BI.webm

Completed lab

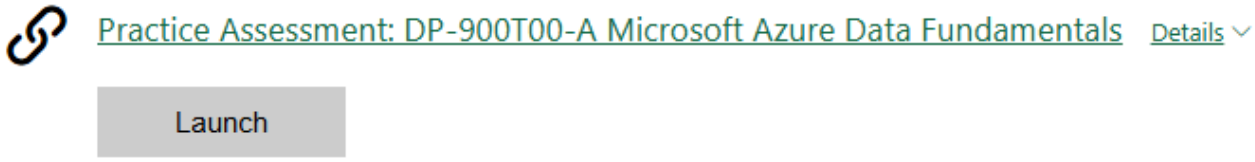




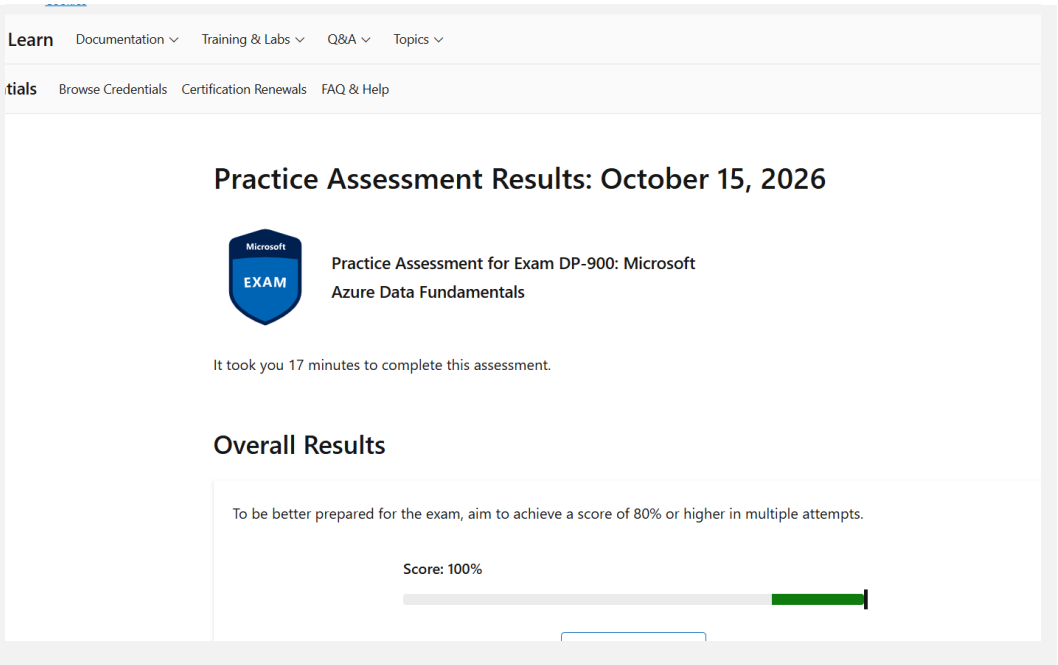
Day 4: Task 2: Skillable (Optional)

After completing all the labs, you can complete the Azure DP-900 practice exam and paste your result below – this is open book and please research and discuss your answers as a team.

1) Visit https://leeptalent.learnondemand.net/
2) Sign into your account
3) Select activity **DP -900T00-A Introduction to Microsoft Azure Data**
4) Find and Launch **Practice Assessment: DP-900T00-A Microsoft Azure Data Fundamentals**



Completed lab



Day 4: Task 3 (Optional)

1. Scenario Background

"Paws & Whiskers" is a growing pet shop that aims to improve its business by analysing sales, customer information, and inventory data. Currently, the data is collected manually or stored in spreadsheets. Management is interested in transitioning to Microsoft Azure to streamline data storage, analysis, and reporting, enabling them to make data-driven decisions.

2. Data Laws and Regulations

Identify and explain the data laws and regulations relevant to handling customer data within the proposal. Ensure you cover the following points:

- **GDPR Compliance**: Highlight the importance of adhering to the General Data Protection Regulation (GDPR), particularly as it relates to storing and processing customer information.
- **Data Protection Act (DPA) 2018**: Outline how the DPA 2018 may affect the way "Paws & Whiskers" collects and stores data, ensuring compliance with UK laws on data privacy.
- **Other Industry Standards**: Research any additional data protection standards or regulations that may apply to pet shop data, particularly if they involve sensitive or payment information.

3. Azure Service Recommendations

Recommend Microsoft Azure services that would suit the company's data analysis needs and explain why these services are suitable. Your recommendations should include:

- **Data Storage**: Identify suitable storage options, such as **Azure Blob Storage** or **Azure SQL Database**, and discuss the benefits of each for storing large datasets, including inventory, sales transactions, and customer details.

- **Data Analysis Tools**: Recommend tools such as **Azure Machine Learning** for customer behaviour analysis or **Azure Synapse Analytics** for analysing sales trends.
- **Data Integration and Automation**: Explain how services like **Azure Data Factory** could automate data collection and integration processes, improving efficiency.

4. Data Types and Data Modelling

Define the types of data "Paws & Whiskers" will need to work with and describe your approach to data modelling:

- **Data Categories**: Identify key data types, such as customer demographics, transaction history, pet inventory, and product categories.
- **Data Modelling Approach**: Outline how you would structure this data using a relational model or a data warehouse approach, considering factors like tables, entities, relationships, and primary keys.

5. Data Storage Formats and Structures in Azure

Discuss how you would store data within Azure and the formats you would recommend:

- **Data Formats**: Specify recommended formats (e.g., CSV for raw data imports, JSON for structured data, Parquet for analytics) and explain why these formats are suitable for specific data types.
- **Data Security and Encryption**: Include recommendations for securing data using Azure's built-in encryption features and access controls to ensure compliance with data privacy regulations.

6. Additional Considerations

Provide any other considerations that might enhance data handling and efficiency in Azure, such as:

- **Backup and Disaster Recovery**: Outline a backup plan using **Azure Backup** or **Azure Site Recovery** to safeguard against data loss.

- **Data Visualisation**: Discuss potential use of **Power BI** within Azure for creating dashboards that provide management with real-time insights into sales and customer trends.
- **Future Scalability**: Comment on how Azure services can scale as the business grows, accommodating larger datasets and more complex analyses.

Submission Guidelines:

1. **Structure**: Ensure your report is well-organised, with sections for each task (e.g., Data Laws, Azure Services, Data Types, etc.).
2. **Formatting**: Include headings, bullet points where appropriate, and any visuals or diagrams that support your explanations.
3. **References**: Cite any resources or regulations referenced in the report.
4. **Length**: Aim for 1500-2000 words.

Proposal for Improving Data Management at Paws & Whiskers Using Microsoft Azure

1. Introduction

Paws & Whiskers is a growing pet shop that wants to improve how it manages and analyses its business data. Currently, the company collects information manually or stores it in spreadsheets. Although this may work for a smaller business, it can become difficult to manage as the company grows. For example, spreadsheets may contain duplicate information, human errors or outdated stock records.

Microsoft Azure is a cloud computing platform that provides services for storing, managing and analysing data. By moving to Azure, Paws & Whiskers could improve data accuracy, automate repetitive tasks and make better business decisions.

This proposal outlines the relevant data protection laws, recommended Azure services, data modelling approaches, storage formats and security measures. It also considers how the company could use its data to understand customer behaviour, monitor sales and manage inventory more efficiently.

2. Data Laws and Regulations

Protecting customer information is important because Paws & Whiskers may collect personal details such as names, email addresses, telephone numbers, home addresses and purchase histories.

UK GDPR

The UK General Data Protection Regulation (UK GDPR) sets out requirements for how businesses collect, use, store and protect personal information.

Paws & Whiskers should follow key principles, including:

- Lawfulness, fairness and transparency: Customers should understand why their information is collected and how it will be used.
- Data minimisation: Only collect information necessary for business purposes.
- Accuracy: Ensure customer details are correct and updated when needed.
- Storage limitation: Avoid keeping personal information longer than necessary.
- Security: Protect data against unauthorised access, loss or misuse.

For example, if customers provide email addresses to receive order confirmations, the company should not automatically use these addresses for marketing without meeting the relevant marketing rules.

Customers also have rights, including the right to access their personal information and request corrections or deletion in certain circumstances.

Data Protection Act 2018

The Data Protection Act 2018 works alongside the UK GDPR and provides additional rules governing personal data in the UK.

Paws & Whiskers should implement suitable procedures to ensure customer information is handled securely. This includes restricting access to authorised employees, maintaining accurate records and establishing a process for responding to data protection requests.

For example, if a customer asks what information the company holds about them, Paws & Whiskers should be able to locate and provide the relevant information within the applicable legal timeframe.

Other Relevant Standards

Payment Card Industry Data Security Standard (PCI DSS): If Paws & Whiskers accepts card payments, it should ensure payment information is processed securely. Using a trusted payment provider can reduce the need to store sensitive card details directly.

Privacy and Electronic Communications Regulations (PECR): These rules are relevant if the business sends promotional emails or text messages. Marketing communications must meet applicable consent and opt-out requirements.
Overall, following these regulations would help protect customers, maintain trust and reduce the risk of data breaches or financial penalties.

3. Azure Service Recommendations

Microsoft Azure provides several services that could help Paws & Whiskers manage its growing data.

Data Storage

Azure SQL Database

Azure SQL Database is a managed relational database service suitable for storing structured information such as customer records, sales transactions and inventory.
Its benefits include:

- Organising information into tables.
- Allowing employees to retrieve information using SQL queries.
- Supporting relationships between customers, products and sales.
- Providing built-in security and automated backup capabilities.

For example, management could use SQL queries to identify the five best-selling pet food products each month.

Azure Blob Storage

Azure Blob Storage is suitable for storing large amounts of files and unstructured data, including product images, exported spreadsheets, CSV files and historical sales data.
It is useful because it offers flexible storage capacity and different storage tiers, helping the business manage costs.

Data Analysis Tools

Azure Synapse Analytics

Azure Synapse Analytics combines data integration and analytical capabilities. Paws & Whiskers could use it to analyse sales trends across different months, compare product categories and identify seasonal purchasing patterns.
For example, the business could investigate whether dog food sales increase during certain periods or which product categories generate the most revenue.

Azure Machine Learning

Azure Machine Learning could be introduced as the business develops and collects more data.
It could help the company:

- Predict which products customers may purchase.
- Forecast future demand for popular products.
- Identify purchasing patterns.
- Support personalised product recommendations.

However, machine learning may not be necessary immediately, as simpler SQL and Power BI analysis could meet the company's initial needs.

Data Integration and Automation

Azure Data Factory

Azure Data Factory can automate the movement and transformation of data between different sources.
For example, instead of manually combining daily sales spreadsheets, Paws & Whiskers could create a scheduled process that imports sales files, checks the data and transfers it into Azure SQL Database.
This would reduce manual work, improve consistency and allow employees to spend more time analysing information rather than organising spreadsheets.

4. Data Types and Data Modelling

Paws & Whiskers will need to organise different types of data to understand its customers, sales and stock levels.

Data categories:

- Customer data: Customer ID, name, age range, contact details.- Understanding customer demographics.
- Sales data: Transaction ID, date, quantity, total price.- Monitoring revenue and sales trends.
- Inventory data: Product ID, stock quantity, supplier.-Managing stock levels.
- Product data: Product name, category, price.-Identifying popular products.
- Pet information: Pet type, breed, age.-Understanding customer preferences.

Customer and transaction information would mainly be structured data, while product images and documents would be unstructured data.

Data Modelling Approach

A relational database would be suitable because it organises information into separate but connected tables.

For example, Paws & Whiskers could create:

- Customers table: CustomerID (primary key), Name, Email.
- Products table: ProductID (primary key), ProductName, CategoryID, Price.
- Sales table: SaleID (primary key), CustomerID, SaleDate.
- SaleItems table: SaleItemID (primary key), SaleID, ProductID, Quantity.
- Inventory table: InventoryID (primary key), ProductID, StockQuantity.

A primary key uniquely identifies each record, while a foreign key connects records across tables.

For example, CustomerID in the Sales table would link each transaction to the correct customer. The SaleItems table would allow one transaction to contain multiple products. This structure reduces duplicate information, improves data accuracy and makes SQL queries easier to perform.

As the business grows, a data warehouse could also be introduced to store historical sales information for more detailed analysis.

5. Data Storage Formats and Structures in Azure

Choosing suitable file formats would help Paws & Whiskers store and process its information efficiently.

Recommended Data Formats

CSV (Comma-Separated Values)

CSV files would be useful when transferring existing spreadsheet data into Azure. For example, the company could export its weekly sales records as CSV files before importing them into a database.

CSV files are simple, widely supported and easy to work with in Excel and other data analysis tools.

JSON (JavaScript Object Notation)

JSON would be suitable for exchanging information between different applications, particularly online ordering systems.

For example, a customer's online order could be represented in JSON, containing the order number, products purchased and delivery information.

Parquet

Parquet is a column-based file format designed for efficient storage and analysis of large datasets.

Paws & Whiskers could use Parquet to store historical sales data for analysis in Azure Synapse Analytics. Its compression capabilities can reduce storage requirements and improve analytical performance.

Data Security and Encryption

Protecting customer information should be a priority when transferring data to Azure. The company should implement several security measures:

- Encryption at rest: Use Azure Storage encryption and Transparent Data Encryption to protect stored information.
- Encryption in transit: Use secure HTTPS and TLS connections when transferring data.
- Role-based access control (RBAC): Restrict access according to employees' responsibilities.
- Multi-factor authentication (MFA): Add another layer of protection when employees sign in.
- Azure Key Vault: Securely manage passwords, secrets and encryption keys.

For example, a sales assistant may need access to stock availability but should not automatically have access to every customer's personal information.

Microsoft documents these security features for Azure SQL Database and Azure data factory (msmbaldwin, 2026).

6. Additional Considerations

Backup and Disaster Recovery

Paws & Whiskers should have a backup and recovery plan to prevent data loss caused by technical failures, accidental deletion or cyberattacks.

Azure SQL Database provides automated backups and point-in-time recovery, allowing the company to restore its database to an earlier state within the configured retention period.

For additional protection, the company could:

- Use storage redundancy for important files.
- Configure suitable backup retention periods.
- Regularly test whether data can be restored.
- Monitor systems for failures and security threats.

Azure Backup could be considered for other supported workloads, while Azure Site Recovery would be more relevant if the business later introduced virtual machines requiring disaster recovery (msmbaldwin, 2026).

Data Visualisation Using Power BI

Power BI could help management understand business performance through interactive reports and dashboards.

For example, a dashboard could display:

- Total daily, weekly and monthly sales.
- Best-selling products.
- Revenue by product category.
- Products running low on stock.
- Customer purchasing patterns.

These dashboards could be connected to Azure SQL Database and refreshed regularly, reducing the need to prepare reports manually. Depending on the connection method, some reports could provide near-real-time information.

This would help management make quicker decisions about ordering products, planning promotions and managing inventory.

Future Scalability and Cost

One advantage of Azure is its ability to support growing businesses.

As Paws & Whiskers gains more customers and products, it could increase its storage and computing resources without purchasing additional physical servers.
However, the company should also monitor costs. Introducing several advanced services immediately may be unnecessary for a small business.
A sensible approach would be to begin with Azure SQL Database, Blob Storage and Power BI, then introduce more advanced analytical services when needed.

7. Conclusion

Moving to Microsoft Azure could help Paws & Whiskers improve how it stores, manages and analyses its business information.
Azure SQL Database would provide a structured way to manage customers, sales and inventory, while Blob Storage would support files and historical data. Azure Data Factory could reduce manual processes, and Power BI would allow management to identify trends through interactive dashboards.
The company must also follow UK data protection legislation and implement appropriate security measures to protect customer information.
Overall, a gradual move to Azure would help Paws & Whiskers become more efficient, improve decision-making and prepare for future business growth.

Word count =1721

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Additional Resources

1) DataCamp Recommended Courses
   A. Understanding Microsoft Azure - DataCamp Learn
   B. Introduction to Azure - DataCamp Learn
   C. Understanding Cloud Computing - DataCamp Learn
   D. Introduction to Microsoft Fabric - DataCamp Learn

2) Introduction to Microsoft Azure Data: In this course, students will gain foundational knowledge of core data concepts and related Microsoft Azure data services.
   Course DP-900T00-A: Introduction to Microsoft Azure Data - Training | Microsoft Learn

3) Explore how to implement data analytics solutions on a single platform with Microsoft Fabric. Integrate, transform, and store data to train AI models and create insightful reports.
   Get started with Microsoft Fabric - Training | Microsoft Learn

4) Azure Data Fundamentals Flashcards
   Azure Data Fundamentals Flashcards | Quizlet

5) Learn the Fundamentals of Microsoft Fabric in 38 minutes

Learn the Fundamentals of Microsoft Fabric in 38 minutes

6) View expert-led demos to quickly learn how to get started and accomplish your goals with Azure services
Watch Azure Demos to Get Started | Microsoft Azure

7) Practice Assessment for Exam DP-900: Microsoft Azure Data Fundamentals
Practice Assessment | Microsoft Learn

Course Notes

It is recommended to take notes from the course, use the space below to do so.

END OF WORKBOOK

Please check through your work thoroughly before submitting and update the table of contents if required.

Please send your completed work booklet to your trainer.